

Project Design Phase-II

Data Flow Diagram & User Stories

Date	28.09.2026
Team ID	SWTID-2026-4440
Project Name	Auto Ticket Classification using FLOW DESIGNER
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts the system requirements graphically, demonstrating how ticket data enters the system, how Flow Designer triggers and processes classification rules, how dependent choice values are verified, and how the records are updated and assigned in the database.

DFD Level 0 (Context Level Diagram)

Level 0 DFD illustrates the overall boundary of the Auto Ticket Classification system, interacting with external entities (Caller / School User and IT Support Groups) and the core Ticket database.

DFD Level 0: Auto Ticket Classification using FLOW DESIGNER

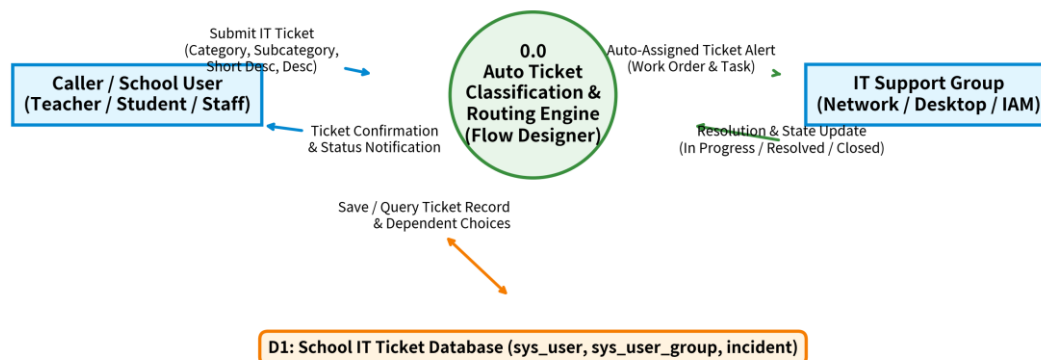


Figure 1: DFD Level 0: Context Diagram for Auto Ticket Classification Engine

DFD Level 1 (Decomposition Diagram)

Level 1 DFD decomposes the system into detailed sub-processes: Ticket Submission & Field Validation (1.0), Category & Subcategory Dependency Logic (2.0), Flow Designer Auto-Classification & Routing (3.0), and Notification & State Lifecycle Management (4.0).

DFD Level 1: Sub-Processes of Auto Ticket Classification

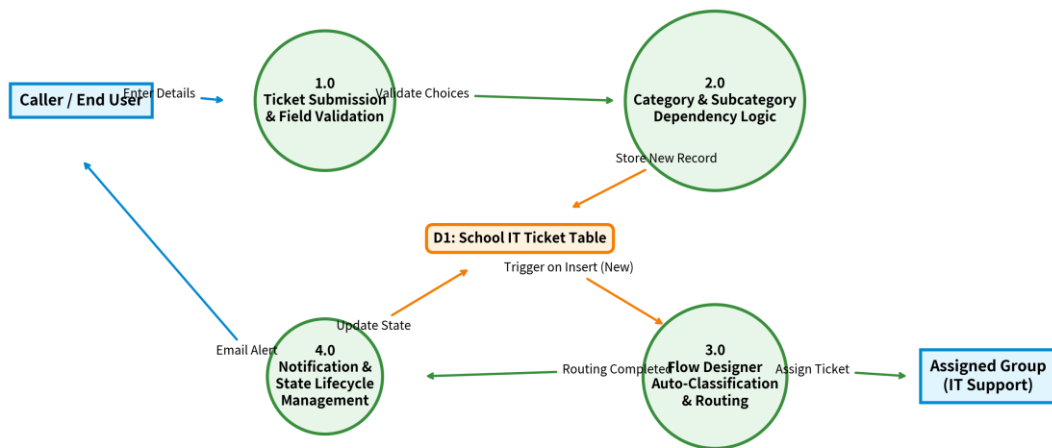


Figure 2: DFD Level 1: Sub-Processes & Data Stores for Automated Routing

Choice Field Configuration & Dependency Architecture

To ensure structured and consistent data capture, the School IT Ticket table contains dependent dropdown fields. The Subcategory choices are dynamically populated based on the selected Category choice values as shown below:

Category to Subcategory Mapping Rules:

- Network → Wi-Fi
- Hardware → Projector
- Access → Forgot password
- Performance → Slow computer

Dictionary Entry & Dependent Field Setup (Backend Design)

In the Dictionary Entry for Subcategory ('sys_choice'), the 'Use dependent field' checkbox is checked and mapped to the 'Category' field. Each choice value carries the corresponding dependent value (e.g., access, network, performance, hardware) as illustrated in the system screenshot:

- Now scroll down to the choice section.
- Then add the dependent fields for which option dependent on the category gives the category choice backend name to the subcategory choice values.
- Wi-Fi → Network.
- Projector → Hardware.
- Forgot password → Access.
- Slow computer → Performance.

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	False	2026-02-06 21:45:02	access
Wi-Fi	wi-fi	en	0	False	2026-02-06 21:44:50	network
Slow Computer	slow computer	en	3	False	2026-02-06 21:45:15	performance
Projector	projector	en	1	False	2026-02-06 21:45:28	hardware

- Then save the Form.
- Then it will display each choice to each one separately based on the category subcategory displays.

Figure 3: Dictionary Entry: Subcategory Choices with Dependent Values

Frontend Form Design & Dynamic Dropdown Verification

When a user selects a Category on the Ticket Form, the Subcategory dropdown immediately filters to display only the relevant sub-options:

The screenshot shows the 'Incident Workflow' 'New record' form. The 'Number' field contains 'INC00909'. The 'Category' dropdown is set to 'Network'. The 'Subcategory' dropdown is set to 'Wi-Fi', which is highlighted in blue. The 'State' dropdown is set to 'None'. The 'Assigned group' and 'Assigned to' fields are empty. The 'Short description' field is empty. The 'Description' field is empty. A 'Submit' button is at the bottom left.

Figure 4: Ticket Form: Network Category showing dependent Wi-Fi Subcategory

The screenshot shows the 'Incident Workflow' 'New record' form. The 'Number' field contains 'INC00909'. The 'Category' dropdown is set to 'Access'. The 'Subcategory' dropdown is set to 'Wi-Fi', which is highlighted in blue. The 'State' dropdown is set to 'None'. The 'Assigned group' and 'Assigned to' fields are empty. The 'Short description' field is empty. The 'Description' field is empty. A 'Submit' button is at the bottom left.

The screenshot shows the 'Incident Workflow' 'New record' form. The 'Number' field contains 'INC00909'. The 'Category' dropdown is set to 'Performance'. The 'Subcategory' dropdown is set to 'Wi-Fi', which is highlighted in blue. The 'State' dropdown is set to 'None'. The 'Assigned group' and 'Assigned to' fields are empty. The 'Short description' field is empty. The 'Description' field is empty. A 'Submit' button is at the bottom left.

The screenshot shows the 'Incident Workflow' 'New record' form. The 'Number' field contains 'INC00909'. The 'Category' dropdown is set to 'Hardware'. The 'Subcategory' dropdown is set to 'Wi-Fi', which is highlighted in blue. The 'State' dropdown is set to 'None'. The 'Assigned group' and 'Assigned to' fields are empty. The 'Short description' field is empty. The 'Description' field is empty. A 'Submit' button is at the bottom left.

Figure 5: Ticket Form: Verification of Hardware, Access, and Performance Dependent Choices

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Caller (School Staff / Student)	Ticket Submission & Fields	USN-1	As a Caller, I want to submit a ticket with Number, Caller, Short Description, and Description so that IT support is aware of my issue.	All mandatory fields are captured and a unique ticket number is generated.	High	Sprint-1
Caller (School Staff / Student)	Dependent Choices Setup	USN-2	As a Caller, I want Subcategory choices (Wi-Fi, Projector, Forgot password, Slow computer) to filter automatically based on Category (Network, Hardware, Access, Performance).	Subcategory dropdown displays only options dependent on selected Category.	High	Sprint-1
System (Flow Designer)	Auto Classification & Routing	USN-3	As an automated Flow Designer, I want to trigger when a new ticket is submitted and classify it into the correct Support Group based on Category/Subcategory.	Tickets with Network/Wi-Fi route to Network Group; Hardware/Projector route to Desktop Support; Access to IAM; Performance to Systems Team.	High	Sprint-1
IT Support Technician	Assignment & State Management	USN-4	As an IT Support Technician, I want tickets assigned to my group with State set to 'In progress' so that I can immediately start resolution.	Assigned Group field is auto-populated and State transitions from 'New' to 'In progress'.	High	Sprint-1
Caller (School Staff / Student)	Automated Notifications	USN-5	As a Caller, I want to receive email notifications when my ticket is created, assigned, and updated.	Automated emails are sent with ticket number, category, and assigned technician details.	Medium	Sprint-1
IT Support Technician	Resolution & Closure	USN-6	As an IT Support Technician, I want to update the ticket state to 'Resolved' or 'Closed' with	Ticket state updates properly and caller is	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			resolution notes upon fixing the issue.	notified of resolution.		
IT Administrator	Workflow Monitoring & Analytics	USN-7	As an IT Administrator, I want to view Flow Designer execution logs and classification metrics to ensure optimal routing performance.	Execution history and routing success rates are viewable in Flow Designer dashboard.	Medium	Sprint-2